

Homework solutions for 02YELMA

Chapter: Special Theory of Relativity

1. Classically, one would simply add the velocities:

$$u_{\text{cl}} = 50 + 10 = 60 \text{ km/h.}$$

The correct relativistic velocity addition formula is

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}},$$

where

$$v = 50 \text{ km/h}, \quad u' = 10 \text{ km/h.}$$

Thus

$$u = \frac{60}{1 + \frac{(50)(10)}{c^2}}.$$

Since

$$\frac{u'v}{c^2} \ll 1,$$

we use the series expansion

$$\frac{1}{1+x} \approx 1-x \quad (|x| \ll 1).$$

Taking

$$x = \frac{u'v}{c^2},$$

we get

$$u \approx (u' + v) \left(1 - \frac{u'v}{c^2} \right).$$

Therefore

$$u \approx 60 \left(1 - \frac{500}{c^2} \right).$$

The classical error is

$$\Delta u = u_{\text{cl}} - u \approx 60 \cdot \frac{500}{c^2}.$$

Hence the relative error is

$$\frac{\Delta u}{u} \approx \frac{500}{c^2}.$$

Using

$$c = 3 \times 10^8 \text{ m/s} = 1.08 \times 10^9 \text{ km/h,}$$

we obtain

$$\frac{500}{c^2} = \frac{500}{(1.08 \times 10^9)^2} \approx 4.3 \times 10^{-16}.$$

Thus the percentage error is

$$\boxed{\% \text{ error} \approx 4.3 \times 10^{-14} \%}$$

which is essentially zero, so for everyday speeds simple addition is completely sufficient.

2. (a) Without Special Relativity

Muon lifetime:

$$t_0 = 2.2 \times 10^{-6} \text{ s}$$

Speed:

$$v = 0.998c$$

Distance traveled:

$$d = vt_0 = (0.998)(3.0 \times 10^8)(2.2 \times 10^{-6})$$

$$d \approx 6.59 \times 10^2 \text{ m}$$

$$d \approx 659 \text{ m}$$

Since

$$659 \text{ m} \ll 15\,000 \text{ m},$$

the muons would not reach the Earth's surface.

(b) With Special Relativity

Time dilation:

$$t = \frac{t_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

Since

$$\frac{v}{c} = 0.998,$$

we have

$$1 - (0.998)^2 = 1 - 0.996004 = 0.003996$$

Thus

$$\sqrt{0.003996} \approx 0.0632$$

Therefore

$$t = \frac{2.2 \times 10^{-6}}{0.0632} \approx 3.48 \times 10^{-5} \text{ s}$$

So the dilated lifetime is

$$t \approx 3.5 \times 10^{-5} \text{ s}.$$

(c) Comparison and Conclusion

Using the dilated lifetime,

$$d = vt = (0.998)(3.0 \times 10^8)(3.48 \times 10^{-5})$$

$$d \approx 1.04 \times 10^4 \text{ m}$$

$$d \approx 10.4 \text{ km}$$

This is much larger than the non-relativistic result and is comparable to the atmospheric thickness but it is still smaller than 15 km:

$$10.4 \text{ km} \lesssim 15 \text{ km}$$

The reason we can still detect them at the Earth's surface is that the muon decay is a random process and some muons will survive longer than the average lifetime. Therefore many muons can still survive long enough to reach the Earth's surface. Without special relativity they would decay too quickly, so time dilation is necessary to explain the observed muons at ground level.

3. a)

$$\vec{E}' = \begin{pmatrix} 0 \\ \gamma E_0 \\ 0 \end{pmatrix} \quad \vec{B}' = \begin{pmatrix} 0 \\ 0 \\ -\frac{\gamma v}{c^2} E_0 \end{pmatrix}$$

b) S' observes an electric field in y' direction with magnitude γE_0 which is scaled by the Lorentz factor γ . Since $\gamma \geq 1$, $|\vec{E}'| \geq |\vec{E}|$. Even though S frame observes no magnetic field, S' frame observes magnetic field in z' direction with magnitude $\frac{\gamma v}{c^2} E_0$.

c)

$$\vec{u}' = \begin{pmatrix} u'_x \\ u'_y \\ u'_z \end{pmatrix} \Rightarrow \vec{u}' \times \vec{B}' = \begin{pmatrix} u'_y B'_z \\ -u'_x B'_z \\ 0 \end{pmatrix}$$
$$\vec{F}' = q\vec{E}' + q\vec{u}' \times \vec{B}' = \begin{pmatrix} qu'_y B'_z \\ q\gamma E_0 - qu'_x B'_z \\ 0 \end{pmatrix}$$

d) Equations of motion:

- **x-direction:** $m\dot{u}'_x = qu'_y B'_z$
- **y-direction:** $m\dot{u}'_y = -qu'_x B'_z + q\gamma E_0$
- **z-direction:** $m\dot{u}'_z = 0$

where $B'_z \equiv -\frac{\gamma v}{c^2} E_0$ and the motion is coupled in x and y directions. The particle experiences no force and therefore no acceleration in the z direction, so there is no motion in the z direction. To decouple the differential equations, we differentiate the first, and then, insert the second:

$$\ddot{u}'_x = \frac{q}{m} \dot{u}'_y B'_z = \frac{q}{m} \left[-\frac{q}{m} u'_x B'_z + \frac{q}{m} \gamma E_0 \right] B'_z$$

$$\begin{aligned}
&= -\frac{q^2}{m^2} B_z'^2 u'_x + \frac{q^2}{m^2} \gamma E_0 B'_z \\
&= -\omega^2 u'_x + \frac{q^2}{m^2} \gamma E_0 B'_z, \quad \text{where } \omega \equiv \frac{q B'_z}{m}
\end{aligned}$$

$$\ddot{u}'_x + \omega^2 u'_x = \left(\frac{q^2}{m^2} \gamma E_0 B'_z \right) \rightarrow \ddot{u}'_x + \omega^2 u'_x = C$$

This is a 2nd order inhomogenous differential equation and its general solution is $u'_x(t') = y_c(t') + y_p(t')$, where $y_c(t')$ is the complementary solution and $y_p(t')$ is the particular solution.

$$y_c(t') = A \cos(\omega t') + B \sin(\omega t')$$

To find $y_p(t')$, observe that the inhomogenous part is a constant;

$$C = \frac{q^2}{m^2} \gamma E_0 B'_z$$

A convenient solution suggestion would be constant: $y_p(t') = \text{const} = y_p$

Plug const. into $\ddot{u}'_x + \omega^2 u'_x = C \Rightarrow 0 + \omega^2 y_p = C \Rightarrow y_p = C/\omega^2$

$$y_p = \frac{\frac{q^2}{m^2} \gamma E_0 B'_z}{\omega^2} = \frac{\gamma E_0}{B'_z}$$

General solution:

$$u'_x(t') = A \cos(\omega t') + B \sin(\omega t') + \frac{\gamma E_0}{B'_z}$$

Consider the initial conditions to determine A and B:

$$u'_x(0) = A \cdot 1 + \frac{\gamma E_0}{B'_z} = 0 \Rightarrow A = -\frac{\gamma E_0}{B'_z}$$

$$\dot{u}'_x(t') = -A\omega \sin(\omega t') + B\omega \cos(\omega t')$$

$$\dot{u}'_x(0) = B\omega : \text{acceleration at } t = 0 \text{ in x direction.}$$

Since at $t' = 0$,

$$\vec{F}' = \begin{pmatrix} 0 \\ q\gamma E_0 \\ 0 \end{pmatrix}$$

no force and therefore no acc. in x direction.

$$\dot{u}'_x(0) = B\omega = 0 \Rightarrow B = 0$$

$$u'_x(t') = \frac{\gamma E_0}{B'_z} (1 - \cos(\omega t'))$$

Time evolution of velocity in x-direction.

Solution for $u'_y(t')$

$$\begin{aligned} \dot{u}'_y &= -\frac{q}{m}u'_xB'_z + \frac{q}{m}\gamma E_0 = -\frac{q}{m}\left[\frac{\gamma E_0}{B'_z}(1 - \cos(\omega t'))\right]B'_z + \frac{q}{m}\gamma E_0 \\ &= -\frac{q}{m}\gamma E_0 + \frac{q}{m}\gamma E_0 \cos(\omega t') + \frac{q}{m}\gamma E_0 = \frac{q}{m}\gamma E_0 \cos(\omega t') \end{aligned}$$

$$\frac{d[u'_y]}{dt} = \frac{q}{m}\gamma E_0 \cos(\omega t') \Rightarrow u'_y(t') = \frac{q}{m}\gamma E_0 \frac{1}{\omega} \sin(\omega t') + C$$

Impose initial conditions $u'_y(0) = 0 = C \Rightarrow u'_y(t') = \frac{q}{m}\gamma E_0 \frac{1}{\omega} \sin(\omega t')$

$$u'_y(t') = \frac{q}{m}\gamma E_0 \frac{1}{\frac{qB'_z}{m}} \sin(\omega t') = \frac{\gamma E_0}{B'_z} \sin(\omega t')$$

Summary:

$$u'_x(t') = \frac{\gamma E_0}{B'_z} [1 - \cos(\omega t')]$$

$$u'_y(t') = \frac{\gamma E_0}{B'_z} \sin(\omega t')$$

Time dependence of velocities describe a CYCLOIDAL MOTION in $x'y'$ plane.

CYCLOID: It is the path traced by a point on the circumference of a circle as it rolls along a straight line without slipping. (Imagine drawing a chalk dot on a bicycle tire and then rolling the bicycle forward the dot traces a cycloid.) (See figures below for visualization.)

$$x(\theta) = R\theta - R\sin\theta = R(\theta - \sin\theta)$$

$$y(\theta) = R - R\cos\theta = R(1 - \cos\theta)$$

where $\theta = \omega t$ and $R = \frac{\gamma E_0}{B'_z}$

$$\dot{x}(\theta) = R - R\cos\theta = R(1 - \cos\theta)$$

$$\dot{y}(\theta) = R\sin\theta$$

Summary:

$$u_x'(t') = \frac{\gamma E_0}{B_z'} [1 - \cos(\omega t')]$$

$$u_y'(t') = \frac{\gamma E_0}{B_z'} \sin(\omega t')$$

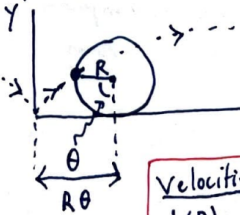
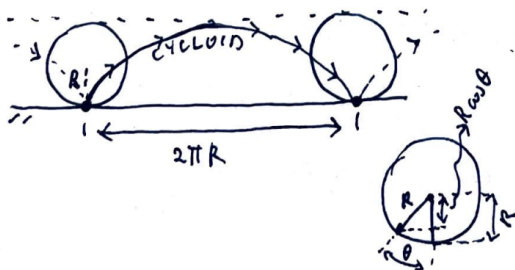
$$u_z'(t') = 0$$

Time dependence of velocities describe a CYCLOIDAL MOTION in $x'y'$ plane.

COMPARE

$$\theta \equiv \omega t' \text{ and } R = \frac{\gamma E_0}{B_z'}$$

CYCLOID: It is the path traced by a point on the circumference of a circle as it rolls along a straight line without slipping. (Imagine drawing a chalk dot on a bicycle tire and then rolling the bicycle forward - the dot traces a cycloid).



$$y(\theta) = R - R \cos(\theta) = R(1 - \cos \theta)$$

$$x(\theta) = R\theta - R \sin \theta = R(\theta - \sin \theta)$$

Velocities:

$$\dot{x}(\theta) = R - R \cos \theta = R(1 - \cos \theta)$$

$$\dot{y}(\theta) = R \sin \theta$$

TRAJECTORY OF THE CHARGE IN S' FRAME

